Code -

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%%writefile vector_add.cu
#include <iostream>
#include <cuda_runtime.h>
using namespace std;

__global__ void addVectors(int* A, int* B, int* C, int n) {
    int i = blockIdx.x * blockDim.x + threadIdx.x;
    if (i < n) C[i] = A[i] + B[i];
}

int main() {
    int n;
    cout << "Enter size of vectors: ";
    cin >> n;

    int size = n * sizeof(int);

    int *A, *B, *C;
    cudaMallocHost(&A, size);
    cudaMallocHost(&B, size);
    C = (int*) malloc(size);

    cout << "Enter elements of vector A: ";
    for(int i=0;i<n;i++) cin >> A[i];

    cout << "Enter elements of vector B: ";
    for(int i=0;i<n;i++) cin >> B[i];

    int *dA, *dB, *dC;
    cudaMalloc(&dA, size);
    cudaMalloc(&dB, size);
    cudaMalloc(&dC, size);

    cudaMemcpy(dA, A, size, cudaMemcpyHostToDevice);
    cudaMemcpy(dB, B, size, cudaMemcpyHostToDevice);

    int blockSize = 256;
    int numBlocks = (n + blockSize - 1) / blockSize;

    addVectors<<<numBlocks, blockSize>>>(dA, dB, dC, n);
    cudaDeviceSynchronize();

    cudaMemcpy(C, dC, size, cudaMemcpyDeviceToHost);

    cout << "Resultant vector C: ";
    for(int i=0;i<n;i++) cout << C[i] << " ";
    cout << endl;

    return 0;
}
```

Output -

```
Enter size of vectors: 5
Enter elements of vector A: 1 2 3 4 5
Enter elements of vector B: 5 4 3 2 1
Resultant vector C: 6 6 6 6 6
```